

TEACH YOURSELF LECTURE

Lecture 01

Computer Systems (HCI)

Chapter lecture — 16 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

Computer Science XII · Bilingual support: English + اردو

کمپیوٹر سائنس — بارہویں جماعت: خود سیکھیں، خود پرکھیں

16 teach-yourself lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XII

How to use this lecture

Learn this chapter **on your own**. Read *Learn it*, try the worked example on paper, then answer *Check yourself* before looking at the answers. The green box repeats the idea in Urdu. ★ Golden topics are high-yield in board exams.

یہ لیکچر خود سیکھیں۔ مثال کاغذ پر حل کریں، پھر جواب دیکھنے سے پہلے خود کو پرکھیں۔

Lectures in this pack

Chapter 1: Computer Systems (HCI). Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

L01. 1.1 Fundamentals of Human-Computer Interaction

L03. 1.2 Applications of HCI (Domains of Life)

L05. Types of User Interaction

L07. 1.4 Importance of HCI

L09. 1.6 Need Analysis for Interface Design

L11. 1.8 Methods for Improving HCI

L13. 1.9.3–1.9.5 Wireframing, Figma and Prototypes

L15. 1.11.2 Testing Methods (Usability, A/B, Automated)

L02. 1.1.1 Traditional Interaction and 1.1.2 Natural Interaction

L04. 1.3 Components of HCI

L06. Interface Types, Environment and Feedback

L08. 1.5 Accessibility Principles

L10. 1.7 Human-Computer Interaction Problems

L12. 1.9 User Interface Design (UI vs UX)

L14. 1.11 Test and Evaluate HCI (Evaluation Methods)

L16. Career Opportunities in UI/UX

CHAPTER 1

Computer Systems (HCI)

This chapter teaches you Human-Computer Interaction (HCI): how people and computers talk to each other, how to design screens that are easy and fair to use, and how to test those designs before they go live. You will apply every idea to things you already use in Pakistan — a bank ATM, JazzCash, and a school portal.

اس باب میں آپ ہیومن کمپیوٹر انٹرایکشن (HCI) سیکھیں گے: انسان اور کمپیوٹر آپس میں کیسے بات کرتے ہیں، اسکرینیں کیسے آسان اور منصفانہ بنائی جاتی ہیں، اور انہیں عوام تک پہنچانے سے پہلے کیسے جانچا جاتا ہے۔

What you will learn

- Define HCI (also called CHI and MMI) and name the five sensory channels of interaction.
- Describe HCI applications in health care, banking, education, and networking.
- Explain environment and the two kinds of feedback.
- Carry out a need analysis before designing an interface.
- Differentiate UI from UX, and compare wireframes, Figma work and prototypes.
- Differentiate traditional interaction from natural interaction with local examples.
- Name the components of HCI and classify types of user interaction and interface.
- State why HCI is important and apply five accessibility principles.
- Identify common HCI problems and methods for improving HCI.
- Choose evaluation and testing methods (SUS, NPS, usability, A/B, automated audits).

1.1 Fundamentals of Human-Computer Interaction

LEARN IT

Human-Computer Interaction (HCI) is the study and design of technology so that it **serves people**. An HCI system is easy to use, efficient, and responsive to human needs. HCI is also known as **CHI** (Computer-Human Interface) or **MMI** (Man-Machine Interaction). The three names mean the same idea: the meeting point of a person and a machine.

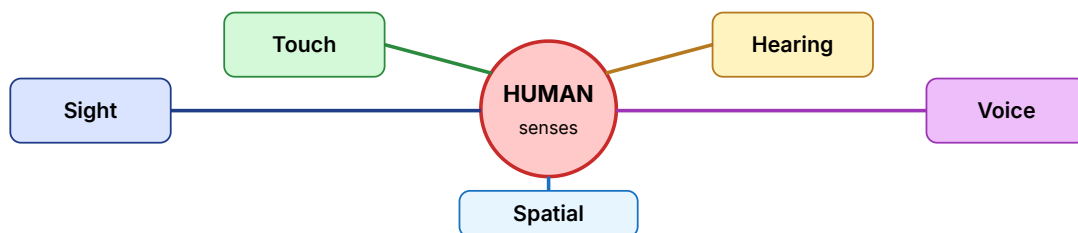
HCI is not only "making a pretty screen". It asks: Can a student in Hyderabad find results on the school portal in 30 seconds? Can a grandmother send JazzCash without help? Can a person standing at an outdoor ATM in bright sunlight read the buttons?

Sensory channels of interaction

Humans and computers communicate through **sensory channels**. A sensory channel is a path that carries information to (or from) one of the human senses. Computers must send and receive signals on the same channels, or the conversation breaks.

Channel	How humans use it	How computers respond
Sight (بصارت)	Look at screens, graphics or videos.	Displays visuals on monitors and screens.
Touch (چھونا)	Tap, type or swipe.	Responds with clicks or vibrations (haptics).
Hearing (سنتا)	Listen to sounds or voices.	Provides audio feedback via speakers.
Voice (آواز)	Speak commands (for example, "Alexa").	Processes speech and replies.
Spatial (مکانی)	Move in physical space.	Tracks position or body movement (for example, VR).

Haptics means physical feedback you can feel, such as a phone vibrating when a JazzCash payment succeeds. **Spatial** interaction means the system notices where your body is, as in a VR headset or a Kinect camera.



Five sensory channels meet at the human. HCI design must support the channels the user actually needs.

WORKED EXAMPLE

Worked example: map a JazzCash payment to channels. Sara in Sukkur sends Rs 500 to her brother.

1. **Sight:** she looks at the JazzCash home screen and the amount field.
2. **Touch:** she taps "Send Money" and types the mobile number.
3. **Voice (optional):** she could say "JazzCash, send five hundred rupees" if voice commands are on.
4. **Hearing + haptics:** a beep and a short vibration confirm "Payment successful".

If the app showed no tick, no sound and no vibration, Sara would not know whether the money left. That is an HCI failure, not a banking failure.

REMEMBER IT

Remember the three names: HCI = CHI = MMI. Think "Human meets Machine." Five channels: **S-T-H-V-S** — Sight, Touch, Hearing, Voice, Spatial. "See The Handy Voice in Space."

COMMON MISTAKE

Mistake: Writing "HCI is only about computer hardware" or mixing up Voice with Hearing.

Fix: HCI is about the *design of the conversation* between people and machines. Hearing is what you *listen to* (speakers). Voice is what you *speak* (microphone + speech software).

اردو میں سمجھیں

ہیومن کمپیوٹر انٹرایکشن (HCI) کا مطلب ہے ٹیکنالوجی کو اس طرح ڈیزائن کرنا کہ وہ انسانوں کے لیے فائدہ مند، آسان اور مؤثر ہو۔ اسے CHI یا MMI بھی کہتے ہیں۔ انسان اور کمپیوٹر پانچ حسی راستوں سے بات کرتے ہیں: بصارت، چھونا، سنا، آواز اور مکانی حرکت۔ ہیپٹکس (Haptics) وہ ارتعاش ہے جو آپ کو چھو کر محسوس ہوتا ہے۔

CHECK YOURSELF

1. Give two other names for HCI.
2. Which channel tracks body movement in VR?
3. Is a phone vibration sight, touch (haptics), or voice?

Answers: 1. CHI (Computer-Human Interface) and MMI (Man-Machine Interaction). 2. Spatial. 3. Touch / haptics.

1.1.1 Traditional Interaction and 1.1.2 Natural Interaction

LEARN IT

1.1.1 Traditional interaction is the basic way people use everyday devices. It relies on standard input and output: a mouse, a keyboard, a keypad, a monitor, speakers, or a simple touchscreen tap and swipe.

- **Computer:** moving a mouse and typing on a keyboard; output via monitor and speakers.
- **Smartphone (traditional use):** tapping and swiping on a touchscreen; output via screen lighting and vibrations.

1.1.2 Natural interaction is advanced, smart technology that lets people interact **without a mouse or keyboard**. It feels more human — you speak, move, or show your face, and the system understands.

- **Voice:** speaking commands into a device (for example, "Alexa, play a song").
- **Body and gesture:** facial recognition to unlock a phone, or playing VR games by waving hands.

Feature	Traditional	Natural
Main input	Mouse, keyboard, keypad, simple tap/swipe	Voice, face, hand/body gesture, movement
Feels like	Operating a machine	Talking or moving as you do with people
Examples	TV remote, ATM keypad, school-portal login typed on a PC	FaceID, Siri, Kinect games, "Hey Google"
Typical output	Screen, speakers, vibration	Spoken reply, unlock, tracking of the body in VR

WORKED EXAMPLE

Worked example: sort today's tools. List three traditional and three natural interfaces you might use in one day in Sindh.

Traditional: (1) HBL ATM keypad to withdraw cash. (2) College computer lab: mouse + keyboard to open the school portal. (3) Television remote at home.

Natural: (1) Unlocking a phone with FaceID. (2) Asking Google Assistant "Call Aslam". (3) A VR science demo in which you pick a molecule by waving your hand.

Notice: the *same smartphone* can be traditional (you type a PIN) or natural (the camera reads your face). The classification depends on **how you interact**, not on the brand of the device.

REMEMBER IT

Test sentence: If you must press keys or tap labelled buttons, it is traditional. If the machine understands speech, face or body movement, it is natural.

COMMON MISTAKE

Mistake: "Touchscreens are always natural interaction."

Fix: Ordinary tapping and swiping on a phone is still listed as traditional in this chapter. Natural interaction starts when the system understands voice, face, or full-body gesture without a mouse or keyboard.

اردو میں سمجھیں

روایتی تعامل (Traditional Interaction) میں ماؤس، کی بورڈ، کی پیڈ یا عام ٹیپ اور سوائپ استعمال ہوتے ہیں، جیسے اے ٹی ایم کی پیڈ۔ قدرتی تعامل (Natural Interaction) میں آواز، چہرہ یا جسم کی حرکت سے کام ہوتا ہے، جیسے فیس آئی ڈی (FaceID) یا "الیکسا، گانا چلاؤ"۔ درجہ بندی آلے کی نہیں، تعامل کے طریقے کی ہوتی ہے۔

CHECK YOURSELF

1. Is an ATM keypad traditional or natural?
2. Give one voice example of natural interaction from the source.
3. Why can one smartphone be both traditional and natural?

Answers: 1. Traditional. 2. "Alexa, play a song" (or any spoken command). 3. Because typing a PIN is traditional, while FaceID or a voice assistant on the same phone is natural — it depends on the method, not the device.

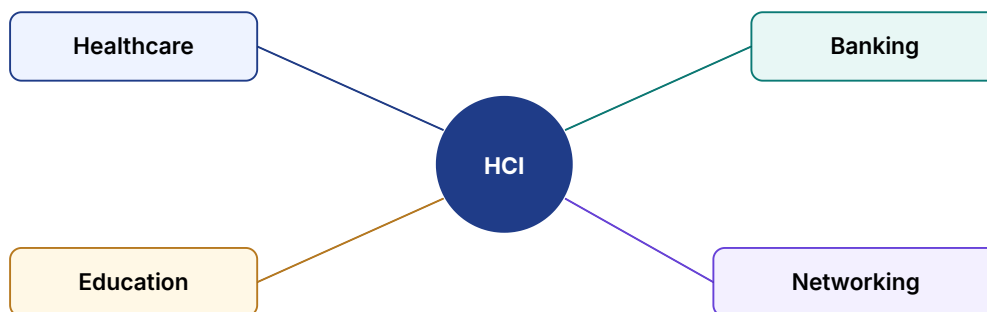
1.2 Applications of HCI (Domains of Life) ★ Golden

LEARN IT

HCI is used to design and develop interactive systems such as **desktop applications, websites and mobile apps**. These are **human-centred solutions** built to meet user needs across many domains of life. A domain is an area of daily activity — health, money, school, talking to people.

When you design for a domain, you design for that domain's **users, tasks and risks**. A mistake on a games app is annoying. A mistake on a medical or banking screen can harm a person or lose money. That is why examiners treat this topic as golden.

Domain	Application examples	User benefit
Health care	Patient portals, telemedicine apps, assistive technologies	Improves access to medical services remotely
Banking	Mobile banking apps, secure ATM interfaces	Enables quick transactions and secure financial management
Education	E-learning platforms (LMS), online exam portals	Enhances teaching, learning and student engagement
Networking and communication	Social media, messaging apps (WhatsApp), Zoom	Connects people globally through collaboration tools



HCI sits at the centre of four everyday domains: healthcare, banking, education and networking.

WORKED EXAMPLE

Worked example: the same HCI idea in three Pakistani systems.

Step 1 — ATM (banking). You insert a card, the screen shows large buttons (Fast Cash, Balance, Other), you enter a PIN on the keypad, and you take cash. HCI benefit: a quick, secure transaction without a human teller.

Step 2 — JazzCash (banking + communication). You log in with a PIN or fingerprint, choose Send Money, type a number and amount, confirm, and see "Successful" with a JazzCash SMS. HCI benefit: money moves in seconds from a shop in Larkana to a student in Karachi.

Step 3 — School portal (education). A Grade XII student opens the college LMS, taps "Download date sheet", and submits a leave form. HCI benefit: learning and admin tasks happen without standing in a queue at the office.

Step 4 — Compare the user benefit. Banking HCI saves time and keeps money safe. Education HCI raises engagement. Health HCI (a Sehat/telemedicine app) brings a doctor to a village phone. Networking HCI (WhatsApp, Zoom) keeps a family in Dubai in touch with parents in Sukkur.

REMEMBER IT

Exam mnemonic H-BEN: Health, Banking, Education, Networking. For each, write: one application + one user benefit. That structure scores full marks on "applications of HCI".

COMMON MISTAKE

Mistake: Listing brand names only ("WhatsApp, JazzCash, Zoom") without saying the *domain* and the *user benefit*.

Fix: Always write three parts: domain → example → what the user gains. "Banking: JazzCash mobile app — enables quick, secure transfers."

ایچ سی آئی کی ایپلیکیشنز صحت، بینکنگ، تعلیم اور نیٹ ورکنگ کے شعبوں میں انسانی ضروریات پوری کرتی ہیں۔ صحت میں مریض پورٹل اور ٹیلی میڈیسن دور سے علاج دیتے ہیں، بینکنگ میں موبائل ایپ اور اے ٹی ایم تیز اور محفوظ لین دین دیتے ہیں، تعلیم میں ای لرننگ اور آن لائن امتحان مشغولیت بڑھاتے ہیں، اور وائس ایپ یا زوم لوگوں کو جوڑتے ہیں۔

EXAM TIP

Very common 5–8 mark question: "Explain applications of HCI in different domains of life with examples." Draw a four-row table (Health care, Banking, Education, Networking & communication). Each row: one or two application names + the user benefit from the source. Add a local line (ATM / JazzCash / school portal) to show you can apply the idea. Do not write a story about only one app.

CHECK YOURSELF

1. Name the four domains of HCI given in the source.
2. What user benefit does HCI bring in banking?
3. Give one education application and one networking application.

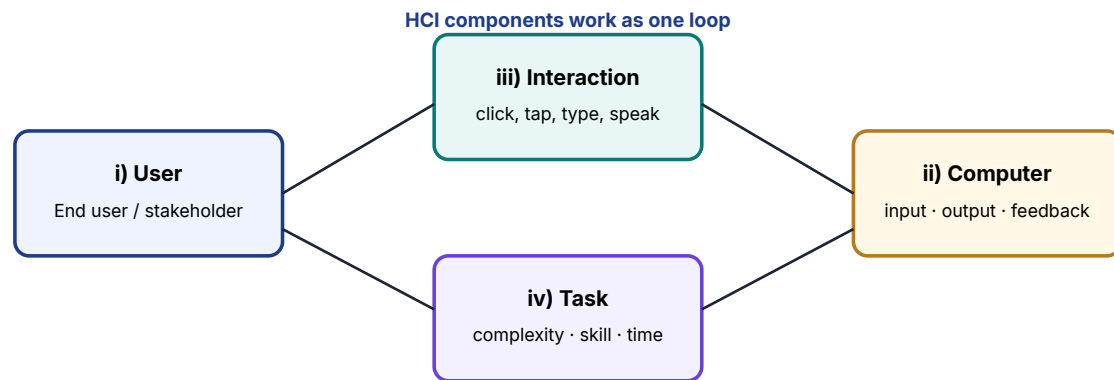
Answers: 1. Health care, banking, education, networking and communication. 2. Quick transactions and secure financial management. 3. Education: LMS or online exam portal. Networking: social media, WhatsApp or Zoom.

1.3 Components of HCI

LEARN IT

HCI relies on several primary components that work together. If one component is weak, the whole interaction feels broken. The first four are the people, the machine, the conversation, and the job to be done.

	Component	Meaning
i	User (صارف)	Any individual or group that interacts with a computer system to achieve defined goals. Two categories: end users (people who directly operate the system, from beginners to experts) and stakeholders (people with an interest in the outcome, such as managers and clients).
ii	Computer system (کمپیوٹر سسٹم)	The platform where humans and computers communicate through input, output and feedback using an interface. It connects human actions with machine responses.
iii	User interaction (صارف کا تعامل)	The process through which a user communicates with a system by performing actions (clicking, tapping, typing, speaking). It transforms user intent into system responses.
iv	Task (کام)	The specific action users perform (for example, booking a ticket). It has three characteristics: task complexity (filling a form vs designing a 3D model), user knowledge (basic typing vs advanced programming), and time requirement (ATM withdrawal vs online passport application).



The user performs a task through interaction with a computer system. All four parts must fit together.

WORKED EXAMPLE

Worked example: label the four components of an HBL ATM withdrawal.

1. **User:** end user = the account holder standing at the machine; stakeholder = the bank that owns the ATM and must keep it safe and working.
2. **Computer system:** the ATM hardware plus the bank's software; it takes PIN input, shows a screen, and dispenses notes (output and feedback).
3. **User interaction:** insert card → type PIN → choose "Cash" → type amount → confirm. Intent ("I want Rs 5,000") becomes a machine response (cash + receipt).
4. **Task:** withdraw cash. Complexity: low (a few buttons). User knowledge: basic (PIN and amounts). Time: about one minute — much shorter than an online passport application.

REMEMBER IT

U-C-I-T: User, Computer system, Interaction, Task. "U Can Interact with Tasks." Then add three task questions: How hard? What skill? How long?

COMMON MISTAKE

Mistake: Calling the bank manager an "end user" of the ATM, or saying a task has only one property (difficulty).

Fix: The manager is usually a **stakeholder**. The person pressing the keys is the **end user**. A task always has complexity, knowledge and time — mention all three in long answers.

اردو میں سمجھیں

ایچ سی آئی کے چار بنیادی اجزاء ہیں: صارف (اینڈ یوزر جو خود سسٹم چلاتا ہے اور اسٹیک ہولڈر جس کا نتیجے میں مفاد ہے)، کمپیوٹر سسٹم (ان پٹ، آؤٹ پٹ اور فیڈبیک کا پلیٹ فارم)، صارف کا تعامل (کلک، ٹیپ، ٹائپ، بولنا) اور کام (ٹاسک) جس کی پیچیدگی، مطلوبہ مہارت اور وقت مختلف ہوتا ہے۔ اے ٹی ایم سے رقم نکالنا آسان اور مختصر کام ہے؛ پاسپورٹ کی آن لائن درخواست لمبا کام ہے۔

CHECK YOURSELF

1. What is the difference between an end user and a stakeholder?
2. Name the three characteristics of a task.
3. In a school portal, who is a stakeholder?

Answers: 1. End users directly operate the system; stakeholders have an interest in the outcome (managers, clients) but may not click the buttons. 2. Task complexity, user knowledge, time requirement. 3. The principal, IT admin or board office — they care about results even if students are the end users.

Types of User Interaction

LEARN IT

User interaction can happen in many different ways depending on the technology used. Learn every type with one clean example — board questions often ask you to "list and explain" five or more.

Interaction type	Description	Examples
Direct manipulation	Handling digital objects on a screen exactly like physical objects.	Dragging files, zooming images with fingers
Command-based	Giving precise instructions through typed text commands.	MS-DOS, Linux terminal (<code>dir</code> , <code>cd ..</code>)
Menu-driven	Selecting from predefined lists or options instead of typing.	Clicking "Start Test" or "Download" on a portal
Form fill	Entering structured data into blank text fields.	Sign-up forms, ticket booking screens
Conversational	Communicating with systems using natural human language (text or voice).	Chatbots, ChatGPT, Google Assistant
Gesture-based	Using body or hand movements to control systems.	Swiping touchscreens, VR hand gestures
Voice interaction	Speaking commands directly to the system.	"Call Aslam" on a smartphone
Context-aware	Systems sense the environment and act automatically without a prompt.	Smart lighting turning on when you enter a room
Biometric	Using biological features for interaction and authentication.	Fingerprint unlock, FaceID
Haptic and tactile	Providing physical feedback to confirm actions.	Phone vibration when typing
Multimodal	Combining two or more interaction types together.	Voice + touch (speak a search, then tap)

WORKED EXAMPLE

Worked example: a full JazzCash "Send Money" session, labelled by type.

1. Open the app with a fingerprint → **biometric**.
2. Home screen shows Send Money, Bill Payment, Load → you tap Send Money → **menu-driven**.
3. Type the receiver's mobile number and the amount → **form fill**.
4. The keypad vibrates on each tap → **haptic**.
5. Optional: say "Send five hundred to Ali" then tap Confirm → **multimodal** (voice + touch).

Same idea on a **school portal**: menu-driven ("Date Sheet", "Fee Challan"), form fill (leave application), conversational (a chatbot "When is the chemistry practical?").

REMEMBER IT

Group them: hands on screen (direct, gesture, haptic) · words (command, conversational, voice) · lists and blanks (menu, form) · automatic/body (context-aware, biometric) · mix (multimodal).

COMMON MISTAKE

Mistake: Treating "voice interaction" and "conversational" as identical, or calling a dropdown menu "command-based".

Fix: Voice is spoken commands only. Conversational includes typed chat (ChatGPT) as well as voice. Command-based means you type precise codes such as `dir`, not clicking a menu.

اردو میں سمجھیں

تعامل کی اقسام میں براہ راست ہیرا پھیری، کمانڈ، مینو، فارم، گفتگو (چیٹ بوٹ)، اشارہ، آواز، سیاق و سباق سے آگاہ نظام، ہایومیٹرک، ہیپٹک اور ملٹی موڈل شامل ہیں۔ جاز کیش میں فنگر پرنٹ ہایومیٹرک ہے، "رقم بھیجیں" مینو ہے، نمبر لکھنا فارم ہے، اور آواز کے ساتھ ٹیپ ملٹی موڈل ہے۔

CHECK YOURSELF

1. Which type is a Linux `cd ..` command?
2. Smart lighting that turns on when you enter a room is which type?
3. Why is speaking a search and then tapping a result called multimodal?

Answers: 1. Command-based. 2. Context-aware. 3. Because it combines two or more types (voice + touch) in one task.

Interface Types, Environment and Feedback

LEARN IT

Components v–vii complete the HCI picture: the look of the controls, the place where you use them, and the signals that tell you "it worked" or "try again".

v) User interface (UI)

The **user interface** is the medium through which humans interact with computer systems.

- **Graphical User Interface (GUI):** visual elements such as windows, icons and menus (Windows OS, Android).
- **Command-Line Interface (CLI):** text-based typing (MS-DOS, Linux terminal).
- **Touch interface:** tapping, swiping, pinching (smartphones, tablets).
- **Voice User Interface (VUI):** spoken commands (smart speakers).
- **Natural User Interface (NUI):** gestures, body movements or immersive environments (VR headsets, Microsoft Kinect).

vi) Environment

The **environment** is the real-world setting where a person uses a computer system. It is shaped by:

- **Physical conditions** — an ATM outdoors in sunlight, heat, noise, rain.
- **Social conditions** — a quiet classroom, a crowded bazaar, privacy at a PIN pad.
- **Technical hardware** — slow college Wi-Fi, a cracked phone screen, an old browser.

vii) Feedback

Feedback is the signal or response given to confirm or guide actions.

System feedback

Response from the computer: a "Loading..." bar, a "Message Sent!" tick, or an error sound.

User feedback

Input from the user to the system: tapping a "Submit" button, giving a 5-star rating.

WORKED EXAMPLE

Worked example: environment and feedback at a roadside ATM in Hyderabad.

Physical: noon sunlight washes out a dim screen — the GUI must use high contrast.

Social: people stand in a queue; the user needs a privacy shield around the PIN pad.

Technical: the printer may be out of paper, so on-screen "Take your cash" plus a beep still has to confirm the transaction.

System feedback: "Please wait", cash coming out, a printed or SMS receipt.

User feedback: the user presses Confirm, or later rates the bank's app 4 stars.

On a **school portal** during board-exam week, hundreds of students on weak Wi-Fi need a visible "Loading results..." bar. Silence after clicking "View marks" is poor system feedback.

REMEMBER IT

GUI vs CLI vs NUI: GUI = pictures you click. CLI = words you type. NUI = body you move. VUI = voice you speak. Touch = fingers on glass.

Feedback rule: **system talks back; user talks in.**

COMMON MISTAKE

Mistake: Mixing up system feedback with user feedback, or saying "environment" means only the software theme (dark mode).

Fix: A tick mark is system feedback. A 5-star rating is user feedback. Environment is the *real world* around the user — sun, noise, hardware — not the colour of the app (though a good UI will adapt to that world).

اردو میں سمجھیں

یوزر انٹرفیس وہ واسطہ ہے جس سے انسان سسٹم سے بات کرتا ہے: GUI تصاویر اور مینو، CLI لکھی کمانڈ، ٹچ، آواز (VUI) اور قدرتی اشارے (NUI)۔ ماحول وہ حقیقی جگہ ہے جہاں سسٹم استعمال ہوتا ہے — دھوپ میں اے ٹی ایم، خاموش کلاس روم، کمزور وائی فائی۔ سسٹم فیڈبیک مشین کا جواب ہے ("بھیج دیا گیا")؛ صارف کا فیڈبیک صارف کا ان پٹ ہے (سمٹ بٹن یا پانچ ستارے)۔

CHECK YOURSELF

1. Name five types of user interface given in the source.
2. Give one physical and one social example of environment.
3. Is a "Message Sent" tick system feedback or user feedback?

Answers: 1. GUI, CLI, Touch, VUI, NUI. 2. Physical: ATM in sunlight. Social: a quiet classroom. 3. System feedback.

1.4 Importance of HCI ★ Golden

LEARN IT

HCI ensures that machines are approachable and usable, turning complex technology into everyday tools. **Good design** makes technology simple and enjoyable (touchscreens). **Poor design** causes confusion and risks (unclear buttons on medical devices).

This is why HCI is not optional decoration. A confusing JazzCash confirm button can send money to the wrong number. A confusing school-portal "Submit" can lock a student out of an online exam. Importance questions in the board exam expect the five implications below, each with a sentence of explanation.

Implication	Description
1. Enhance productivity	Well-designed interfaces help users work faster, make fewer mistakes, and achieve goals efficiently.
2. Promotes inclusivity	Accessibility features (screen readers, voice commands) empower marginalized groups and ensure equal use of technology for everyone.
3. Drives innovation	Emerging tech such as AR/VR, AI and IoT rely strictly on solid HCI principles for people to actually adopt and use them.
4. Economic impact	User-friendly products gain higher adoption rates, customer loyalty and market success, driving profits.
5. Ethical responsibility	Responsible design ensures technology respects user privacy, reduces bias and supports human well-being.

WORKED EXAMPLE

Worked example: good vs poor HCI, same task.

Task: send Rs 2,000 through JazzCash to a parent.

Good HCI: large "Send Money" button, amount shown in words and numbers, last 4 digits of the number repeated, fingerprint confirm, green "Successful" plus SMS. Result: fast, few mistakes, the user trusts the app (productivity + economic loyalty + ethics of clear consent).

Poor HCI: tiny grey buttons, no amount-in-words, no confirm screen, no tick. Result: wrong number, no idea if it worked, user deletes the app and goes back to a bank queue.

Medical contrast from the source: unclear buttons on a medical device are not just ugly — they are dangerous. That is why ethical responsibility sits in this table.

REMEMBER IT

P-I-I-E-E: Productivity, Inclusivity, Innovation, Economic impact, Ethics. Sentence: "Please Include Innovative, Ethical Earnings."

COMMON MISTAKE

Mistake: Writing only "HCI is important because it makes apps look nice."

Fix: Looks are a tiny part of UI. Importance answers must name productivity, inclusivity, innovation, money and ethics. Quote the medical-device warning if you have time.

اردو میں سمجھیں

اچھا ایچ سی آئی پیچیدہ ٹیکنالوجی کو روزمرہ کا آسان اوزار بناتا ہے، برا ڈیزائن الجھن اور خطرہ پیدا کرتا ہے، جیسے طبی آلے پر غیر واضح بٹن۔ پانچ مضمرات: پیداواری بڑھتی ہے، شمولیت آتی ہے، جدت اپنائی جاتی ہے، معیشت مضبوط ہوتی ہے، اور رازداری و انصاف کی اخلاقی ذمہ داری پوری ہوتی ہے۔

EXAM TIP

Typical wording: "Why is HCI important? Discuss its implications." (8 marks) Write a short opening (good design vs poor design) then five numbered points with one example each. Local examples that earn marks: ATM PIN privacy (ethics), screen reader on a school portal (inclusivity), JazzCash one-tap send (productivity), a bank's app beating a rival because it is easier (economic), AR classroom app that students actually use (innovation). Do not skip ethics — examiners look for privacy and well-being.

CHECK YOURSELF

1. Give one example of poor design mentioned in the source.
2. Which implication is about screen readers and equal use?
3. How does HCI create economic impact?

Answers: 1. Unclear buttons on medical devices. 2. Promotes inclusivity. 3. User-friendly products get higher adoption, loyalty and profits.

1.5 Accessibility Principles

LEARN IT

Accessibility principles are HCI guidelines that ensure digital systems are usable by **all** individuals, including people with diverse physical, sensory and cognitive abilities. They remove barriers to create equal access.

Accessibility is not a "special extra". It is how you make sure a student with low vision, a shopkeeper who cannot use a mouse, or a colour-blind driver can still complete the task.

Principle	What it means	Example
1. High contrast (واضح رنگ)	Make text and pictures stand out clearly against the background.	Black text on a white background in a railway app, readable in bright sunlight.
2. Readable fonts (پڑھنے میں آسان تحریر)	Clear, simple fonts with proper size and spacing.	A school website using large Calibri so homework instructions are easy to read.
3. Colour-blind friendly design (رنگوں کی پہچان)	Do not depend <i>only</i> on colours; also use symbols or text labels.	A traffic app showing a red circle with a "STOP" label, not just a red colour.
4. Inclusive navigation (آسان نیویگیشن)	Give users different ways to move around a system.	A website that lets you use the Tab key to jump between buttons if you cannot use a mouse.
5. Multimodal detection (متعدد حسی ذرائع)	Present information in different formats (sight, sound, touch).	A video with text captions, or a phone vibrating to alert a user.

WORKED EXAMPLE

Worked example: audit a school portal against the five principles.

1. **Contrast:** are result PDFs pale grey on white? Fail — use dark text.
2. **Fonts:** is the fee-challan note in 9-point decorative script? Fail — large Calibri (or a similar simple font), as the source recommends.
3. **Colour:** is "fail" only a red row with no word "Fail"? Fail — add the word, for colour-blind students.
4. **Navigation:** can a student who cannot use a mouse reach "Download date sheet" with Tab + Enter? If not, fail.
5. **Multimodal:** does the principal's video assembly have captions, and does a fee-due alert vibrate as well as flash? If not, deaf or busy students miss it.

The same five checks apply to an outdoor ATM (sunlight contrast) and to JazzCash (large numerals, Urdu labels, vibration on success).

REMEMBER IT

C-F-C-N-M: Contrast, Fonts, Colour-blind, Navigation, Multimodal. "Can Friends Catch New Messages?"

COMMON MISTAKE

Mistake: "Accessibility is only for people with disabilities, so our class portal does not need it."

Fix: The source says usable by **all** individuals. High contrast also helps in sunlight; captions help in a noisy bus; large fonts help tired eyes during board exams.

اردو میں سمجھیں

رسائی پذیری (Accessibility) کے اصول یقینی بناتے ہیں کہ ہر انسان سسٹم استعمال کر سکے، بشمول مختلف جسمانی، حسی اور ذہنی صلاحیتوں والے افراد۔ پانچ اصول: اونچا تضاد (کنٹراسٹ)، پڑھنے میں آسان فونٹ، صرف رنگ پر انحصار نہ کرنا، کی بورڈ سمیت متبادل نیویگیشن، اور معلومات کو نظر، آواز اور لمس کئی شکلوں میں دینا۔

CHECK YOURSELF

1. Why is a red-only "STOP" sign a problem?
2. Which principle does the Tab key support?
3. Give the source's school-website font example.

Answers: 1. Colour-blind users may not see the meaning if there is no symbol or "STOP" text. 2. Inclusive navigation. 3. Large Calibri on a school website for homework instructions.

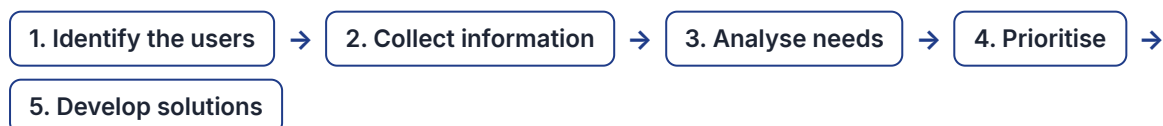
1.6 Need Analysis for Interface Design

LEARN IT

Need analysis is the process of identifying the gap between the **existing situation** and the **desired outcome**. In HCI it is done **before** designing a UI, so that you understand users' goals, tasks, abilities and expectations.

If you skip need analysis, you guess. Guessing is how a school portal ends up with 40 menus and no "Download date sheet" button on the home screen.

Steps in need analysis:



1. **Identify the users:** who exactly will use the system? (elderly people, teenagers, ...)
2. **Information collection:** gather data using interviews, questionnaires or observations.
3. **Analyse user needs:** identify the core problems, goals and requirements.
4. **Prioritise needs:** decide which features are the most critical to build first.
5. **Develop solutions:** design a system that meets those verified needs.

WORKED EXAMPLE

Case study from the source: mobile banking app. Before building the app, the team interviews customers and discovers:

- Users want secure login via Fingerprint / FaceID.
- Elderly users need large buttons and simple navigation.
- Users want support for both English and local languages.

Result: the UI is designed around these proven needs, which makes success more likely.

Apply the five steps to a college portal in Sindh.

1. Users: Grade XI–XII students, parents, clerks, teachers. Not "everyone in Pakistan".
2. Collect: interview 20 students; questionnaire to parents; watch a queue at the office on fee day.
3. Analyse: students cannot find date sheets; parents cannot read English-only fee notices; clerks re-type the same form.
4. Prioritise: (1) date sheet on home screen, (2) Urdu/Sindhi fee notice, (3) online leave form. Fancy animations wait.
5. Solution: a simple portal with those three features, large buttons, bilingual labels.

REMEMBER IT

Need analysis = gap finding. Write: Existing = what hurts now. Desired = what "done" looks like. The UI must close that gap, not decorate it.

COMMON MISTAKE

Mistake: Starting in Figma on day one, then "asking users later".

Fix: Need analysis comes **before** UI design. Features you did not hear from users are guesses. Guessed features waste time and fail in exams when the question is "why do we analyse needs?"

اردو میں سمجھیں

نیڈ اینالیسس موجودہ صورتحال اور مطلوبہ نتیجے کے فرق کو یو آئی بنانے سے پہلے تلاش کرتا ہے۔ پانچ قدم: صارفین کی پہچان، انٹرویو یا مشاہدے سے معلومات، ضروریات کا تجزیہ، اہم خصوصیات کو ترجیح، پھر تصدیق شدہ ضرورتوں پر حل۔ موبائل بینکنگ کی مثال: فنگر پرنٹ لاگ ان، بڑے بٹن، اور انگریزی کے ساتھ مقامی زبان۔

CHECK YOURSELF

1. When in the design process is need analysis done?
2. List the five steps in order.
3. Name three needs found in the mobile-banking case study.

Answers: 1. Before designing the UI. 2. Identify users; collect information; analyse needs; prioritise; develop solutions. 3. Fingerprint/FaceID login; large buttons and simple navigation for elderly users; English plus local languages.

1.7 Human-Computer Interaction Problems ★ Golden

LEARN IT

HCI problems are common difficulties that stop users from using technology smoothly. They happen when systems fail to meet human needs or cognitive limits. **Cognitive limits** means the mind can only hold and process so much at once — if an app asks you to remember three SMS codes at the same time, it has gone past that limit.

Board answers score well when you name the problem, explain it in one line, and give a concrete example (the source examples are safe; a matching local example is even better).

Problem	Explanation and example
Interface complexity	Too many options or a confusing layout. Example: a shopping app with so many menus you cannot find the checkout button.
Poor feedback	The system does not show if an action worked or failed. Example: clicking Submit and the screen freezes with no confirmation.
Inconsistency	Different parts of the app look or act differently. Example: a magnifying-glass icon for Search on one page, a binocular icon on another.
Accessibility gaps	Design ignores people with disabilities. Example: a video tutorial platform without captions for deaf users.
Compatibility issues	Interface fails on different devices or browsers. Example: a website that works on a laptop but crashes on a mobile browser.
Contextual neglect	The system does not adjust to the user's situation. Example: a map app with no dark mode, blinding the driver at night.
High cognitive load	Users must memorize too much or follow overly complex steps. Example: a registration system forcing you to remember three different SMS codes at once.
Trust deficiency	The system feels unsafe or hides security cues. Example: an online store asking for credit cards but lacking an SSL/HTTPS lock icon.

WORKED EXAMPLE

Worked example: diagnose three local systems.

ATM: six tiny unmarked buttons beside the screen and no "Cash is coming" message → interface complexity + poor feedback. A user walks away while cash is still coming out.

JazzCash: "Send" is green on the home screen but orange on the confirm screen, and there is no padlock on the payment page → inconsistency + trust deficiency.

School portal: works in Chrome on a lab PC, blank on a student's Android browser; exam instructions are a caption-free video; login needs a portal password, an email OTP and an SMS OTP on the same screen → compatibility, accessibility gap, high cognitive load.

Once you can name the problem, Section 1.8 tells you how to fix it.

REMEMBER IT

Eight problems in two families: (1) Screen problems — complexity, inconsistency, accessibility, compatibility, context. (2) Mind-and-trust problems — poor feedback, cognitive load, trust. If the user is lost, look at the screen family. If the user is scared or exhausted, look at the mind family.

COMMON MISTAKE

Mistake: Writing "the app is bad" as an HCI problem, or mixing up cognitive load with compatibility.

Fix: Use the exact names from the table. Cognitive load is "too much to remember". Compatibility is "breaks on another device or browser".

ایچ سی آئی مسائل وہ رکاوٹیں ہیں جو سسٹم کو انسانی ضرورت یا ذہنی حد کے مطابق نہ ہونے پر پیدا ہوتی ہیں۔ آٹھ مسائل: انٹرفیس کی پیچیدگی، کمزور فیڈبیک، عدم مطابقت (ایک جیسا نہ ہونا)، رسائی کی کمی، آلہ/براؤزر سے ناسازگاری، سیاق کو نظرانداز کرنا، زیادہ ذہنی بوجھ، اور اعتماد کی کمی (جیسے HTTPS تالے کے بغیر کارڈ مانگنا)۔

EXAM TIP

Common 8-mark question: "Describe problems of HCI with examples." Aim for at least six of the eight, each in the pattern Name → one-line meaning → example. Memorise the source examples (checkout buried in menus, frozen Submit, binocular vs magnifying glass, no captions, laptop-only site, no dark mode, three SMS codes, missing HTTPS lock). Adding ATM / JazzCash / portal examples shows application and often gains the last marks.

CHECK YOURSELF

1. Which problem is a missing HTTPS lock on a payment page?
2. Why are three SMS codes at once an HCI problem?
3. A site works on a laptop but crashes on a phone. Name the problem.

Answers: 1. Trust deficiency. 2. High cognitive load — users must memorize too much. 3. Compatibility issues.

1.8 Methods for Improving HCI

LEARN IT

Once you can name an HCI problem, you fix it with a matching method. The source lists eight methods. Learn each with its tiny example — they pair almost one-to-one with Section 1.7.

- **Enhance usability:** simplify navigation (a clear "Buy Now" button).
- **Strengthen accessibility:** support disabilities (captions on videos).
- **Ensure consistency:** keep layouts uniform (the same "Save" button everywhere).
- **Provide clear feedback:** show action results ("Payment Successful" popup).
- **Improve adaptability:** responsive design (the website scales to mobile screens).
- **Apply user-centred design (UCD):** focus on real user goals and pilot-test.
- **Use multimodal interaction:** allow touch, voice or gesture inputs combined.
- **Continuous testing:** regularly update based on user feedback.

User-centred design (UCD) means the real user's goals drive every decision, and you trial (pilot-test) the design with those users instead of guessing in an office.

Responsive design means the layout rearranges itself for phone, tablet and PC, instead of staying a laptop-only page.

WORKED EXAMPLE

Worked example: match each JazzCash / portal problem to a fix.

Problem (1.7)	Method (1.8)
Cannot find "Send Money"	Enhance usability — one clear button
No captions on help video	Strengthen accessibility
Search icon changes from page to page	Ensure consistency
Submit freezes with no message	Clear feedback — "Payment Successful"
Portal crashes on mobile	Improve adaptability — responsive layout
Built features nobody asked for	UCD — interview, then pilot-test
Elderly user cannot type a long number	Multimodal — voice + touch
Complaints ignored after launch	Continuous testing and updates

REMEMBER IT

Think "problem → method" as a pair. Complexity ↔ usability. No captions ↔ accessibility. Different icons ↔ consistency. Freeze ↔ feedback. Phone crash ↔ adaptability. Guessing ↔ UCD. One input only ↔ multimodal. Launch and forget ↔ continuous testing.

COMMON MISTAKE

Mistake: Naming methods without saying *what they change* ("we will improve HCI" is not an answer).

Fix: Always attach the source example: "Provide clear feedback, for example a Payment Successful popup."

اردو میں سمجھیں

ایچ سی آئی بہتر بنانے کے طریقے: نیویگیشن آسان کرنا، رسائی بڑھانا (کیپشن)، ایک جیسا لے آؤٹ، واضح فیڈبیک ("ادائیگی کامیاب")، ریسپانسیو ڈیزائن، صارف مرکز ڈیزائن (UCD) اور پائلٹ ٹیسٹ، ملٹی موڈل ان پٹ، اور مسلسل جانچ۔ ہر مسئلے کا ایک جوڑی دار حل یاد رکھیں۔

CHECK YOURSELF

1. What does UCD stand for, and what extra step does it include?
2. Which method stops a website from failing on a phone?
3. Give the source's example of clear feedback.

Answers: 1. User-Centred Design; focus on real user goals and pilot-test. 2. Improve adaptability (responsive design). 3. A "Payment Successful" popup.

1.9 User Interface Design (UI vs UX) ★ Golden

LEARN IT

UI design is the process of creating and organizing the **visual elements** of a software application so users can interact with it easily and efficiently.

1.9.1 Components of a user interface

Component	Role
Buttons	Clickable elements that perform actions (Submit, Cancel).
Menus	Lists of options for navigating the system.
Icons	Small symbols representing tools or actions.
Text fields	Boxes where users type information (passwords, search).
Checkboxes / radio buttons	Selecting options in a form. Checkboxes allow several choices; radio buttons allow only one in a group.
Sliders	Adjusting values along a bar (volume, brightness).

1.9.2 User experience (UX)

UX is the overall experience a user has while interacting with a system — how smooth, comfortable and satisfying it feels.

UI is what you see; UX is how it feels.

WORKED EXAMPLE

Source example: food-delivery app. The UI is the nice pictures of burgers and the red buttons. The UX is that the app loads fast, tracking works, and the food arrives without the user getting frustrated.

Local walk-through: JazzCash.

- **UI:** green Send button, JazzCash logo, amount text field, fingerprint icon, menu of Bill / Load / Send.
- **UX:** login in two seconds, no crash in a weak-signal bazaar, money arrives, SMS comes, the user feels safe and finished — or, if it hangs with no tick, the UX is bad even if the buttons look pretty.

School portal: UI = logo, "Login" button, date-sheet menu icon. UX = a student finds the chemistry practical date in under a minute on a phone, without calling the clerk.

Pretty UI + slow, confusing flow = poor UX. Plain UI + fast, clear flow = good UX.

REMEMBER IT

Clothes vs day out. UI is the outfit (colours, buttons, icons). UX is the whole day (Did you arrive on time? Did you get wet? Were you stressed?). Examiners love the line: *UI is what you see; UX is how it feels.*

COMMON MISTAKE

Mistake: Writing "UI and UX are the same" or "UX is just colours".

Fix: UI = visual components (buttons, menus, icons, fields, checkboxes, sliders). UX = overall smoothness, comfort and satisfaction, including speed and whether the task succeeds.

اردو میں سمجھیں

یو آئی ڈیزائن بصری عناصر — بٹن، مینو، آئیکن، ٹیکسٹ فیلڈ، چیک باکس/ریڈیو، سلائیڈر — کو ترتیب دینا ہے۔ یو ایکس پورا تجربہ ہے: کتنا ہموار، آرام دہ اور اطمینان بخش لگا۔ یو آئی وہ ہے جو آپ دیکھتے ہیں؛ یو ایکس وہ ہے جو آپ محسوس کرتے ہیں۔ کھانے کی ڈیلیوری ایپ میں برگر کی خوبصورت تصویر یو آئی ہے، تیز لوڈنگ اور وقت پر کھانا یو ایکس ہے۔

EXAM TIP

Highest-yield short question in the chapter: "Differentiate between UI and UX with an example."

Write: (1) the two definitions, (2) the sentence "UI is what you see; UX is how it feels", (3) the food-delivery example from the source, (4) optionally JazzCash or the school portal. For 5 marks also list four UI components. Do not spend the answer describing Figma — that is the next topic.

CHECK YOURSELF

1. List six UI components from the source.
2. In the food-delivery example, what is UX?
3. Complete: "UI is what you see; UX is ____."

Answers: 1. Buttons, menus, icons, text fields, checkboxes/radio buttons, sliders. 2. The app loads fast, tracking works, food arrives without frustration. 3. how it feels.

1.9.3–1.9.5 Wireframing, Figma and Prototypes

LEARN IT

1.9.3 Wireframing is creating a basic visual outline or blueprint of a user interface **before** the final design. It plans the structure without focusing on colours or detailed graphics.

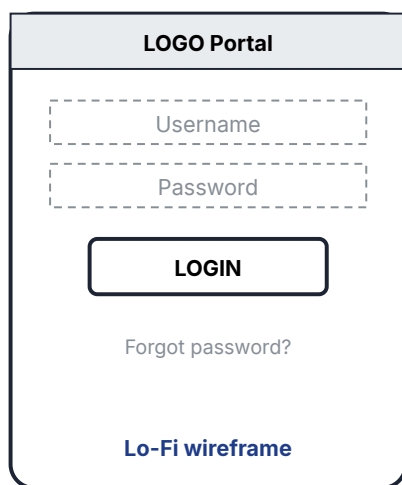
Low-fidelity wireframes (Lo-Fi)

A simple black-and-white sketch showing basic layout (boxes and lines). Used early to test and share ideas quickly. Focuses purely on structure and flow.

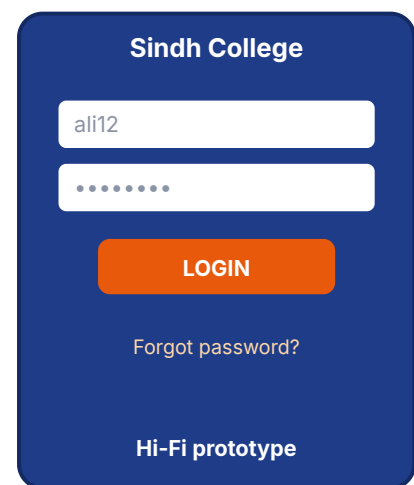
High-fidelity prototypes (Hi-Fi)

A detailed version that looks almost exactly like the final product. Includes real colours, fonts, images and interactive clickable buttons. Used for final user testing.

Lo-Fi sketch → Add details → Hi-Fi prototype



→
add colour



A school-portal login: Lo-Fi boxes and labels on the left; a coloured, realistic Hi-Fi prototype on the right.

1.10 What is Figma?

Figma is a popular, free, web-based tool used to design and collaborate on UI and prototypes. It works directly in a web browser — you do not have to install a heavy program on the college lab PCs.

Creating a low-fidelity design in Figma

1. **Sign up:** access the Figma workspace online.
2. **Create frames:** select the Frame tool (for example an iPhone-sized frame) to act as your screen canvas.
3. **Add shapes:** draw basic rectangles for buttons and circles for avatars.
4. **Add text:** use the text tool to label buttons (for example, "Sign Up").
5. **Duplicate and edit:** copy the frame to create the next screen (for example, the Home Screen).

1.9.5 Prototype design

A **prototype** is an early interactive model used to test design ideas **before writing actual code**. In Figma you use the Prototype tab to link screens together.

- Set a **Trigger** (for example, "On Tap").
- Set an **Action** (for example, "Navigate to Home Screen").

Real users can then click through the fake app to see if the navigation makes sense.

WORKED EXAMPLE

Worked example: JazzCash-style "Send Money" in Figma, without writing code.

1. Lo-Fi on paper or in Figma: three frames — Home (boxes: Send, Bills, Load), Amount (text field + Next), Success (tick + Done).
2. Show the sketch to two classmates. If they cannot find Send, move that box to the top. This is the point of Lo-Fi: cheap changes.
3. In Figma, add a phone frame, rectangles, and labels. Duplicate the frame for Home, Amount and Success.
4. Prototype tab: Trigger = On Tap of the Send rectangle; Action = Navigate to Amount. On Tap of Next → Success.
5. Hand the laptop to a student. If they tap Bills by mistake, the navigation is still wrong — fix it before any programmer writes Java or Kotlin.

REMEMBER IT

Lo-Fi = boxes. Hi-Fi = almost the real app. Prototype = clickable. Figma can hold all three stages. Trigger = what the user does. Action = where the fake app goes.

COMMON MISTAKE

Mistake: Colouring a Lo-Fi wireframe, or thinking a Figma prototype is the finished coded app.

Fix: Lo-Fi stays black-and-white on purpose so people discuss structure, not taste in colours. A prototype tests ideas before code; developers still have to build the real system afterwards.

اردو میں سمجھیں

وائر فریم یو آئی کا بنیادی خاکہ ہے، رنگوں سے پہلے۔ لو فائی سیاہ و سفید ڈیے ہیں؛ ہائی فائی تقریباً حتمی شکل ہے، رنگ، فونٹ اور کلک ہونے والے بٹن کے ساتھ۔ فگما (Figma) مفت ویب ٹول ہے: فریم بناؤ، شکل اور متن ڈالو، فریم کاپی کرو۔ پروٹوٹائپ کوڈ سے پہلے کا چلنے والا نمونہ ہے جس میں ٹرگر (On Tap) اور ایکشن (اگلی اسکرین) جوڑے جاتے ہیں۔

CHECK YOURSELF

1. What does a Lo-Fi wireframe deliberately leave out?
2. Where does Figma run?
3. In Figma prototyping, what is a trigger? Give the source example.

Answers: 1. Colours and detailed graphics (it is structure only). 2. Directly in a web browser. 3. The user event that starts a move between screens, e.g. "On Tap".

1.11 Test and Evaluate HCI (Evaluation Methods)

LEARN IT

Testing checks if the product works properly (finding bugs). **Evaluation** determines whether it meets users' needs (usability). Together they improve the product before release.

A school portal can be "bug-free" (testing passed) and still unusable if students cannot find the date sheet (evaluation failed). You need both.

1.11.1 Evaluation methods

Method	Description
SUS (System Usability Scale)	A standard 10-item questionnaire used to calculate a score for how usable a system is.
NPS (Net Promoter Score)	Asks users: "How likely are you to recommend this app to a friend?" Measures user loyalty.
Formative evaluation	Done during development to collect early feedback and improve the design continuously.
Summative evaluation	Done at the end of development to measure overall success against the original goals.
Expert reviews	Usability specialists review the app against established design standards (heuristics).

Heuristics are agreed rules of good design (for example "show system status", "be consistent"). An expert review checks the product against those rules without waiting for a huge user survey.

WORKED EXAMPLE

Worked example: when to use each method on a school portal.

1. Week 2 of building: show the Lo-Fi to five students, change the menu. That is **formative** evaluation.
2. A teacher who has studied HCI checks the screens against standard rules (Is feedback visible? Are labels consistent?). That is an **expert review**.
3. After the portal is "finished", 30 students fill the 10 SUS questions. A low score means it is still hard to use, even if there are no crashes.
4. Ask: "On a scale of 0–10, would you recommend this portal to a friend in another college?" That is **NPS** (loyalty).
5. Compare the finished portal with the original goals (date sheet in one tap, bilingual fee notice). That is **summative** evaluation.

REMEMBER IT

Formative = forming it (during). Summative = summing it up (end). SUS = 10 questions on usability. NPS = "Would you recommend it?" Expert = specialist vs heuristics.

COMMON MISTAKE

Mistake: Saying testing and evaluation are the same, or mixing formative with summative.

Fix: Testing = bugs. Evaluation = does it meet human needs. Formative is during; summative is at the end against original goals.

اردو میں سمجھیں

ٹیسٹنگ یہ دیکھتی ہے کہ پروڈکٹ ٹھیک چلتا ہے (بگز)؛ ایویلیویشن یہ دیکھتی ہے کہ صارف کی ضرورت پوری ہوئی۔ SUS دس سوالوں والا استعمال کا اسکور ہے، NPS پوچھتا ہے کہ آپ یہ ایپ دوست کو کتنی تجویز کریں گے۔ فارمیٹو جانچ ترقی کے دوران ہوتی ہے، سمیٹو آخر میں اہداف کے خلاف۔ ماہر جائزہ بیورسٹکس (Heuristics) پر ہوتا ہے۔

CHECK YOURSELF

1. What is the difference between testing and evaluation?
2. How many items are on the SUS questionnaire?
3. When is summative evaluation done?

Answers: 1. Testing finds bugs / checks the product works; evaluation checks it meets users' needs (usability). 2. 10. 3. At the end of development, against the original goals.

1.11.2 Testing Methods (Usability, A/B, Automated) ★ Golden

LEARN IT

Testing involves observing how the system behaves under specific conditions. Learn all seven methods. The three named in the heading — usability, A/B and automated — are the ones examiners ask most, but a complete answer also names the others.

1. **Usability testing:** observing real users while they perform realistic tasks, to measure efficiency and satisfaction. If the user gets stuck, designers know where to fix the UI.
2. **Task-based testing:** assigning a specific task (for example, "Find a product and add it to cart") and strictly measuring **completion time**, **success rate** and **number of errors**.
3. **Automated audits:** software bots scan websites to detect technical issues such as broken links, slow loading speeds or low colour contrast.
4. **A/B testing:** comparing two versions of an interface (A and B). Half the users see A, half see B. The one with more clicks or sign-ups wins.
5. **Compatibility testing:** verifying the system on different browsers (Chrome, Safari), devices (mobiles, tablets, PCs) and operating systems.
6. **Performance and load testing:** simulating high traffic (for example 10,000 users logging in at once) to see if speed drops or the server crashes.
7. **Keyboard and screen-reader walkthroughs:** navigating using *only* the keyboard (Tab/Enter) or assistive technologies, to check accessibility for users with disabilities.

Task-based

"Add to cart"
Time: 12 s
Success: Yes

A/B test

Version A | Version B
half / half users
more clicks wins

Automated audit

Bot scanning...
3 broken links
Low contrast

Three golden testing pictures: timed tasks, A versus B, and a bot that finds broken links and weak contrast.

WORKED EXAMPLE

Worked example: test JazzCash "Send Money" and a school portal with numbers.

Usability testing. Sit beside five students. Task: "Send Rs 100 to this test number." Do not help. Note where they pause. If three people look for Confirm at the bottom and it is at the top, that is your UI fix.

Task-based testing (numbers). Same task for 10 students.

- Success: 8 of 10 finish = success rate $(8 \div 10) \times 100 = 80\%$.
- Times: 12, 14, 11, 40, 13, 12, 15, 50, 14, 12 seconds. The two slow times (40, 50) are the students who got stuck — they also made errors.
- Errors: extra taps, wrong field. Count them. High errors + high time = poor HCI, even if most people eventually succeed.

A/B testing. Version A: green "Send" at the top. Version B: large orange "Send" under the amount. 200 users each. If B gets 140 completed payments and A gets 90, B wins. You keep B.

Automated audit. A bot reports 3 broken "Fee challan" links and low contrast on the portal's grey-on-grey marks table. Fix those before the board-exam rush.

Load testing. On result day, 10,000 students log in at 9:00 a.m. If the server crashes, performance testing failed — the UI never even appears.

Keyboard walkthrough. Unplug the mouse. Can a student Tab to "Download date sheet" and press Enter? If focus jumps randomly, accessibility testing failed.

REMEMBER IT

Task-based three numbers: Time, Success, Errors. **A/B:** two versions, split users, the higher clicks/sign-ups win. **Bot:** links, speed, contrast. **Load:** many users at once. **Keyboard:** Tab and Enter only.

COMMON MISTAKE

Mistake: Calling A/B testing "showing both designs to the same user and asking which is prettier".

Fix: Half the users see only A, half see only B. You measure behaviour (clicks, sign-ups), not a taste vote. Also: an automated audit does not replace watching real users — bots cannot see that a student is confused.

اردو میں سمجھیں

سات ٹیسٹنگ طریقے: یوزیبلٹی (حقیقی صارف کو کام کرتے دیکھنا)، ٹاسک بیسڈ (وقت، کامیابی، غلطیاں ناپنا)، خودکار آڈٹ (بوت ٹوٹے لنک، سست رفتار، کم کنٹراسٹ ڈھونڈنے)، اے/بی ٹیسٹ (آدھا صارف اے، آدھا بی؛ زیادہ کلک والا جیتے)، کمپیٹیبلٹی (براؤزر اور آلہ)، لوڈ ٹیسٹ (مثلاً دس ہزار لاگ ان)، اور صرف کی بورڈ یا اسکرین ریڈر سے چل کر رسائی کی جانچ۔

EXAM TIP

Golden 8-mark question: "Explain testing methods used in HCI." Write seven short paragraphs or a seven-row table. For full marks, define usability testing (observe real users on realistic tasks), give the three task-based measures, describe automated audits with the three bot findings (broken links, slow load, low contrast), explain A/B with the half-and-half rule, then add compatibility, load (10,000 users), and keyboard/screen-reader walkthroughs. If the paper asks only "usability, A/B and automated", go deep on those three and still mention the rest in one closing line.

CHECK YOURSELF

1. Name the three measurements in task-based testing.
2. In A/B testing, who sees version A?
3. Give two issues an automated audit can detect.

Answers: 1. Completion time, success rate, number of errors. 2. Half of the users (the other half see B). 3. Any two of: broken links, slow loading, low colour contrast.

Career Opportunities in UI/UX

LEARN IT

The same HCI skills you practised on ATMs, JazzCash and school portals are real jobs. Each role uses a different slice of this chapter.

Career	What the person does
UX designer	Researches user needs to ensure the app is intuitive and satisfying.
UI designer	Designs the visual layout (colours, typography, buttons) using tools such as Figma.
Interaction designer	Focuses on the flow between screens, animations and gestures.
Usability researcher	Runs SUS/NPS tests and A/B tests with real users to find flaws.
Accessibility specialist	Ensures products meet inclusive standards for people with disabilities.

WORKED EXAMPLE

Worked example: who does which JazzCash task?

- "Interview shopkeepers about sending money" → **UX designer**.
- "Pick the green, the typeface and the Send button in Figma" → **UI designer**.
- "On Tap Send, slide to Confirm, then Success" → **interaction designer**.
- "Run SUS and an A/B test of two Send buttons" → **usability researcher**.
- "Check contrast, captions and Tab order" → **accessibility specialist**.

REMEMBER IT

UI looks; UX asks; interaction connects screens; research measures; accessibility includes everyone.

COMMON MISTAKE

Mistake: "UI designer and UX designer are one job."

Fix: In small teams one person may wear both hats, but the source lists them separately: UX researches needs and satisfaction; UI draws the visual layout in tools like Figma.

اردو میں سمجھیں

اس باب سے ملازمتیں: یو ایکس ڈیزائنر صارف کی ضرورت تحقیق کرتا ہے، یو آئی ڈیزائنر فگما میں رنگ اور بٹن بناتا ہے، انٹریکشن ڈیزائنر اسکرینوں کا بہاؤ سنبھالتا ہے، یوزیبیلٹی ریسرچر SUS، NPS اور اے/بی ٹیسٹ چلاتا ہے، اور ایکسیسیبلٹی ماہر سب کے لیے رسائی یقینی بناتا ہے۔

CHECK YOURSELF

1. Who runs SUS, NPS and A/B tests?
2. Which career uses Figma for colours, type and buttons?

Answers: 1. Usability researcher. 2. UI designer.

Chapter 1 Review

Chapter summary

- **HCI** (also CHI, MMI) designs technology to be easy, efficient and responsive to people, using five sensory channels: sight, touch (haptics), hearing, voice and spatial.
- **Traditional interaction** uses mouse, keyboard, keypad or simple tap/swipe; **natural interaction** uses voice, face or body gesture without a mouse or keyboard.
- HCI applications cover **health care, banking, education, and networking/communication**, each with a clear user benefit (remote care, quick secure money, better learning, global connection).
- Four core components: **user** (end user vs stakeholder), **computer system**, **user interaction**, and **task** (complexity, knowledge, time).
- Interaction types include direct manipulation, command-based, menu-driven, form fill, conversational, gesture, voice, context-aware, biometric, haptic and multimodal.
- Interfaces: GUI, CLI, touch, VUI, NUI. **Environment** is physical, social and technical. **System feedback** is the machine's reply; **user feedback** is what the person sends in.
- HCI matters for productivity, inclusivity, innovation, economic impact and ethical responsibility. Poor design (unclear medical buttons) creates risk.
- Five accessibility principles: high contrast, readable fonts, colour-blind friendly labels, inclusive navigation, multimodal presentation.
- **Need analysis** finds the gap before UI design: identify users, collect data, analyse, prioritise, develop solutions (banking case: FaceID, large buttons, local language).
- Eight HCI problems: complexity, poor feedback, inconsistency, accessibility gaps, compatibility, contextual neglect, high cognitive load, trust deficiency.
- Improve HCI with usability, accessibility, consistency, clear feedback, adaptability (responsive design), UCD and pilot tests, multimodal input, and continuous testing.
- **UI** is the visual components (buttons, menus, icons, fields, checkboxes, sliders); **UX** is how the whole experience feels. UI is what you see; UX is how it feels.
- Lo-Fi wireframes plan structure; Hi-Fi prototypes look real; **Figma** is a free browser tool; prototypes use Trigger (On Tap) and Action (navigate) before code.
- Evaluation (SUS, NPS, formative, summative, expert/heuristic reviews) checks needs; testing (usability, task-based, automated, A/B, compatibility, load, keyboard/screen-reader) checks behaviour under conditions.

Key terms

Term	Meaning	اردو
HCI	Human-Computer Interaction; designing technology that is easy, efficient and responsive to people. Also called CHI or MMI.	انسان اور کمپیوٹر کا باہمی تعامل
Haptics	Physical feedback you can feel, such as a vibration.	لمسی / ارتعاشی جواب
Traditional interaction	Using standard input/output such as mouse, keyboard, keypad or simple tap and swipe.	روایتی تعامل
Natural interaction	Interacting without mouse or keyboard, using voice, face or body gesture.	قدرتی تعامل
End user	A person who directly operates the system.	حتمی صارف
Stakeholder	A person or group with an interest in the system's outcome (manager, client).	فریق مفاد
GUI	Graphical User Interface using windows, icons and menus.	تصویری صارف انٹرفیس
CLI	Command-Line Interface using typed text commands.	کمانڈ لائن انٹرفیس
NUI	Natural User Interface using gestures, body movement or immersive environments (VR, Kinect).	قدرتی صارف انٹرفیس
System feedback	The computer's response that confirms or guides an action.	نظام کا جواب
User feedback	Input the user sends to the system (Submit click, star rating).	صارف کی رائے / ان پٹ
Accessibility	Making digital systems usable by all, including people with diverse abilities.	رسائی پذیری
Need analysis	Finding the gap between the current situation and the desired outcome before designing a UI.	ضروریات کا تجزیہ
Cognitive load	How much the user must remember or process at once.	ذہنی بوجھ
UCD	User-Centred Design: focus on real user goals and pilot-test.	صارف مرکز ڈیزائن
UI	The visual elements of an application (buttons, menus, icons, fields, sliders).	صارف انٹرفیس

Term	Meaning	اردو
UX	The overall smoothness, comfort and satisfaction of using a system.	صارف کا تجربہ
Wireframe	A basic outline of a UI that plans structure without colours or detailed graphics.	انٹرفیس کا خاکہ
Lo-Fi	Low-fidelity sketch of boxes and lines used early to test ideas.	کم تفصیل والا خاکہ
Hi-Fi	High-fidelity prototype that looks almost like the final product.	زیادہ تفصیل والا نمونہ
Prototype	An early interactive model used to test design ideas before writing code.	ابتدائی چلنے والا نمونہ
Figma	A free, web-based tool for UI design and prototypes.	فگما (ویب پر یو آئی ٹول)
SUS	System Usability Scale: a 10-item questionnaire that scores how usable a system is.	نظام کی استعمال پذیری کا پیمانہ
NPS	Net Promoter Score: how likely a user is to recommend the app to a friend.	تجویز کا درجہ (وفاداری)
Formative evaluation	Evaluation during development to improve the design continuously.	دورانِ تعمیر جانچ
Summative evaluation	Evaluation at the end, measured against original goals.	اختتامی جانچ
A/B testing	Showing version A to half the users and version B to the other half; the better-performing version wins.	اے/بی موازنہ ٹیسٹ
Automated audit	Bots scanning for broken links, slow pages or low contrast.	خودکار جانچ
Heuristic	An established design standard used in expert reviews.	مستند ڈیزائن اصول

Practice: multiple choice

1. HCI is also known as:
 - a. CPU and ALU
 - b. CHI or MMI
 - c. HTTP or HTML
 - d. GUI only
2. Which sensory channel tracks body movement in VR?
 - a. Sight
 - b. Hearing
 - c. Spatial
 - d. Voice
3. Unlocking a phone with FaceID is an example of:
 - a. natural interaction
 - b. command-based interaction only
 - c. traditional interaction
 - d. CLI
4. Which domain of HCI includes patient portals and telemedicine?
 - a. Banking
 - b. Education
 - c. Networking
 - d. Health care
5. A bank manager who never uses the ATM but cares that it works is a:
 - a. stakeholder
 - b. end user
 - c. haptic device
 - d. wireframe
6. Typing `dir` in MS-DOS is:
 - a. menu-driven
 - b. command-based
 - c. context-aware
 - d. biometric
7. A "Message Sent!" tick mark is:
 - a. user feedback
 - b. a stakeholder
 - c. need analysis
 - d. system feedback
8. Unclear buttons on a medical device mainly show that poor HCI:
 - a. only affects game apps
 - b. is a colour preference
 - c. causes confusion and risk
 - d. improves load testing
9. Not depending only on colour, but adding a "STOP" label, is:
 - a. colour-blind friendly design
 - b. load testing
 - c. CLI design
 - d. A/B testing
10. Need analysis for a UI should be done:
 - a. after the product is sold
 - b. before designing the UI
 - c. only by programmers
 - d. instead of testing
11. An online store that asks for a credit card but shows no HTTPS lock has:
 - a. high contrast
 - b. natural interaction
 - c. summative evaluation
 - d. trust deficiency
12. "UI is what you see; UX is how it feels" means:
 - a. UI and UX are identical
 - b. UX is only the logo colour
 - c. UI is code and UX is hardware
 - d. UI is visual elements; UX is the overall experience
13. A Lo-Fi wireframe should focus on:
 - a. final photographs and brand colours
 - b. structure and flow using boxes and lines
 - c. server load of 10,000 users
 - d. writing the finished Python code
14. The System Usability Scale (SUS) is:
 - a. a 10-item questionnaire that scores usability
 - b. a bot that finds broken links
 - c. a type of ATM keypad
 - d. a Figma trigger
15. In A/B testing:
 - a. every user sees both versions at once and votes for the prettier one
 - b. only experts review heuristics
 - c. half the users see A, half see B, and the version with more clicks or sign-ups wins
 - d. the server is crashed on purpose

Practice: short questions

1. Define HCI. Give two other names for it and list the five sensory channels.
2. Differentiate traditional interaction from natural interaction with one example of each from daily Pakistani use.
3. State four domains of HCI. For each, give one application and one user benefit.
4. Name the four primary HCI components. Distinguish end user from stakeholder, and list the three characteristics of a task.
5. Define GUI, CLI, VUI and NUI, with one example of each.
6. Write five implications of the importance of HCI.
7. List the five accessibility principles with one example each.
8. What is need analysis? Write its five steps and three findings from the mobile-banking case study.
9. Differentiate UI from UX using the food-delivery example. Name six UI components.
10. What is A/B testing? How is it different from an automated audit?

Practice: long / problem questions

1. Explain the components of HCI (user, computer system, user interaction, task, interface types, environment, feedback). Illustrate every part with a labelled walk-through of a person withdrawing cash from an outdoor ATM in Hyderabad.
2. Describe eight HCI problems with examples. Then match each problem to a method of improving HCI. Use JazzCash or a school portal wherever it makes the example clearer.
3. "UI is what you see; UX is how it feels." Explain UI design, its six components, and UX. Then describe Lo-Fi wireframes, Hi-Fi prototypes, the steps of making a low-fidelity design in Figma, and how a Figma prototype uses a trigger and an action. Sketch a Lo-Fi school-portal login.
4. Differentiate testing from evaluation. Explain SUS, NPS, formative evaluation, summative evaluation and expert reviews. Then explain usability testing, task-based testing (including the three measurements), automated audits, A/B testing, compatibility testing, performance/load testing, and keyboard/screen-reader walkthroughs. Apply three of these methods to a college portal on result day.
5. A team is designing a JazzCash-style "Send Money" screen for elderly users in Sukkur who speak Sindhi, stand in bright sunlight, and have never used a CLI. Carry out a need analysis (five steps), list accessibility principles you will apply, choose interaction types, and say which evaluation and testing methods you will use before launch and why.

Answer key

Multiple choice

1. b
2. c
3. a
4. d
5. a
6. b
7. d
8. c
9. a
10. b
11. d
12. d
13. b
14. a
15. c

Short questions — model answers

1. HCI is the design of technology to serve people; systems that are easy to use, efficient and responsive. Also called CHI (Computer-Human Interface) and MMI (Man-Machine Interaction). Channels: sight, touch, hearing, voice, spatial.
2. Traditional uses standard I/O (mouse, keyboard, keypad, simple tap/swipe), e.g. an ATM keypad or typing on a school portal. Natural lets people interact without mouse or keyboard, using voice, face or gesture, e.g. FaceID or "Alexa, play a song".
3. Health care: patient portals / telemedicine — remote access to medical services. Banking: mobile banking and ATMs — quick, secure transactions. Education: LMS and exam portals — better teaching, learning and engagement. Networking: WhatsApp, Zoom, social media — global connection and collaboration.
4. User, computer system, user interaction, task. End users directly operate the system; stakeholders (managers, clients) have an interest in the outcome. Task characteristics: complexity, user knowledge, time requirement.
5. GUI: windows, icons, menus (Windows, Android). CLI: typed commands (MS-DOS, Linux). VUI: spoken commands (smart speakers). NUI: gestures, body, immersive VR/Kinect.
6. Enhance productivity; promote inclusivity; drive innovation (AR/VR, AI, IoT need solid HCI); economic impact (adoption, loyalty, profit); ethical responsibility (privacy, less bias, well-being).
7. High contrast (black text on white in a railway app); readable fonts (large Calibri on a school site); colour-blind friendly (STOP label, not colour alone); inclusive navigation (Tab key); multimodal detection (captions or vibration).
8. Need analysis finds the gap between the existing situation and the desired outcome, before UI design. Steps: identify users; collect information (interviews, questionnaires, observation); analyse needs; prioritise; develop solutions. Banking case: Fingerprint/FaceID login; large buttons and simple navigation for elderly users; English and local languages.
9. UI is the visual elements so people can interact easily; UX is the overall smoothness, comfort and satisfaction. Food app: UI = burger pictures and red buttons; UX = fast load, working tracking, food arrives without frustration. Components: buttons, menus, icons, text fields, checkboxes/radio buttons, sliders.
10. A/B testing shows version A to half the users and version B to the other half; the version with more clicks or sign-ups wins. An automated audit uses bots to scan for technical issues such as broken links, slow loading or low contrast; it does not split users between two designs.

Long questions — model answers

- 1. Components through an outdoor ATM.** User: end user = account holder; stakeholder = bank. Computer system: ATM hardware and bank software linking PIN input to cash output. Interaction: insert card, type PIN, choose Cash, confirm — intent becomes a response. Task: withdraw cash; low complexity, basic knowledge, about one minute (contrast with a long passport application). Interface: GUI on screen + keypad (traditional); possibly fingerprint (biometric). Environment: physical sunlight so contrast must be high; social queue so a privacy shield is needed; technical printer or network may fail. System feedback: "Please wait", cash, receipt/SMS. User feedback: pressing Confirm. If any part is weak (glare, no "cash coming" message), HCI fails even if the bank account is correct.
- 2. Problems and methods.** Interface complexity (buried checkout / hidden Send Money) → enhance usability (clear Buy Now / Send). Poor feedback (frozen Submit) → Payment Successful popup. Inconsistency (magnifying glass vs binoculars, or Send changing colour) → same Save/Send control everywhere. Accessibility gaps (no captions) → strengthen accessibility. Compatibility (laptop-only portal) → responsive/adaptability. Contextual neglect (map with no night mode; ATM unreadable in sun) → design for environment. High cognitive load (three SMS codes) → simplify steps. Trust deficiency (no HTTPS lock on JazzCash-style payment) → show security cues. Also apply UCD with a pilot test, multimodal input (voice + touch for elderly users), and continuous testing after launch.
- 3. UI, UX, wireframe, Figma, prototype.** UI design organises visual elements. Six components: buttons (Submit/Cancel), menus, icons, text fields, checkboxes/radio, sliders. UX = overall experience; UI is what you see, UX is how it feels; food-delivery pictures vs fast delivery. Lo-Fi: black-and-white boxes for structure; Hi-Fi: real colour, type, images, clickable controls for final tests. Figma: sign up in the browser; Frame tool for a phone canvas; rectangles and circles; text labels such as Sign Up; duplicate frames for the next screen. Prototype: before code, Prototype tab, Trigger = On Tap, Action = Navigate to Home. Sketch: header "LOGO / Portal", dashed Username and Password boxes, a LOGIN rectangle, "Forgot password?" — the Lo-Fi login in this chapter.
- 4. Test vs evaluate, then methods.** Testing finds bugs / whether it works; evaluation checks whether it meets user needs. SUS: 10-item usability score. NPS: "recommend to a friend?" (loyalty). Formative: during development. Summative: at the end vs original goals. Expert reviews: specialists vs heuristics. Usability testing: watch real users on realistic tasks. Task-based: time, success rate, errors (e.g. Add to cart in 12 s, success yes). Automated audit: bots find broken links, slow load, low contrast. A/B: half/half, more clicks win. Compatibility: Chrome/Safari, phone/tablet/PC, OS. Load: 10,000 logins at once. Keyboard/screen-reader: Tab/Enter and assistive tech. Result-day portal: load-test the 9:00 a.m. rush; usability-test "find chemistry marks"; automated audit for broken result links and grey-on-grey contrast; optional A/B of two home-screen layouts.
- 5. Send Money for elderly users in Sukkur.** Need analysis: (1) users = elderly Sindhi speakers, not teenagers; (2) collect: interviews and watch them at a JazzCash shop agent; (3) analyse: fear of wrong number, cannot read small English, glare on the phone; (4) prioritise: huge Sindhi "رقم بھیجیں", amount in words, confirm screen, high contrast; (5) solution: that UI, not a CLI. Accessibility: high contrast for sunlight, large simple fonts, not colour-only status, Tab/large targets, vibration + spoken confirm (multimodal). Interaction: menu-driven + form fill + biometric + haptic; optional voice. Evaluate formatively on paper then in Figma; expert review against heuristics; SUS and NPS before launch. Test: usability and task-based ("send Rs 100") measuring time/success/errors; A/B two button sizes; compatibility on cheap Androids; keyboard if a kiosk is used; do not skip a confirm screen — trust and feedback are the main risks.

Self-assessment: I can...

- ☐ define HCI, CHI and MMI and name the five sensory channels.
- ☐ describe HCI in health, banking, education and networking.
- ☐ name the types of user interaction and give an example of each.
- ☐ apply the five accessibility principles and carry out a need analysis.
- ☐ differentiate UI from UX and list six UI components.
- ☐ choose evaluation methods (SUS, NPS, formative, summative, expert).
- ☐ tell traditional interaction from natural interaction with local examples.
- ☐ label user, computer system, interaction, task, UI type, environment and feedback on an ATM, JazzCash or portal.
- ☐ explain why HCI is important using the five implications.
- ☐ name eight HCI problems and the methods that fix them.
- ☐ compare Lo-Fi, Hi-Fi, Figma steps and prototype triggers/actions.
- ☐ explain usability, task-based, automated, A/B, compatibility, load and keyboard tests.